

Scalability & Future Plan

Date	6July2026
Team ID	
Project Name	Plugging into the Future : An Exploration of Electricity Consumption Patterns
Maximum Marks	1Mark

Current System Limitations:

S.No	Limitation	Impact	Priority to Address(High/Medium / Low)
1	Basic Dashboard Analytics	Provides limited visualization and reporting features.	Medium
2	Local Deployment	Application can only be accessed on the local machine; no user access.	High
3	SQ Lite Database	Suitable for small-scale applications but not for enterprise-level systems.	Medium
4	Static Machine Learning Model	Model is not automatically retrained with newly collected data.	Medium

Scalability Plan:

S.No	Scalability Aspect	Current State	Proposed Upgrade/Solution
1	Application Deployment	Local Flask Server	Deploy on Render , Railway , Azure , or AWS Cloud
2	Database	SQ Lite Database	Upgrade to MySQL or Postgre SQL
3	Dashboard Analytics	Basic statistical dashboard	Add interactive charts and real-time analytics

4	Security	Basic Login Authentication	Add password encryption, email verification, and role-based access control
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Future Roadmap:

Phase	Planned Feature/ Enhancement	Target Timeline	Expected Impact
Phase1	Cloud Deployment	Week1–2	Makes the application accessible from any where.
Phase2	Advanced User Authentication	Week3–4	Improves application security and user management.
Phase3	Database Upgrade (MySQL / Postgre SQL)	Week5	Supports large-scale data Storage and better performance.
Phase4	Interactive Dash board& Model Retraining	Week6–7	Enhances analytics, visualization , and prediction accuracy.